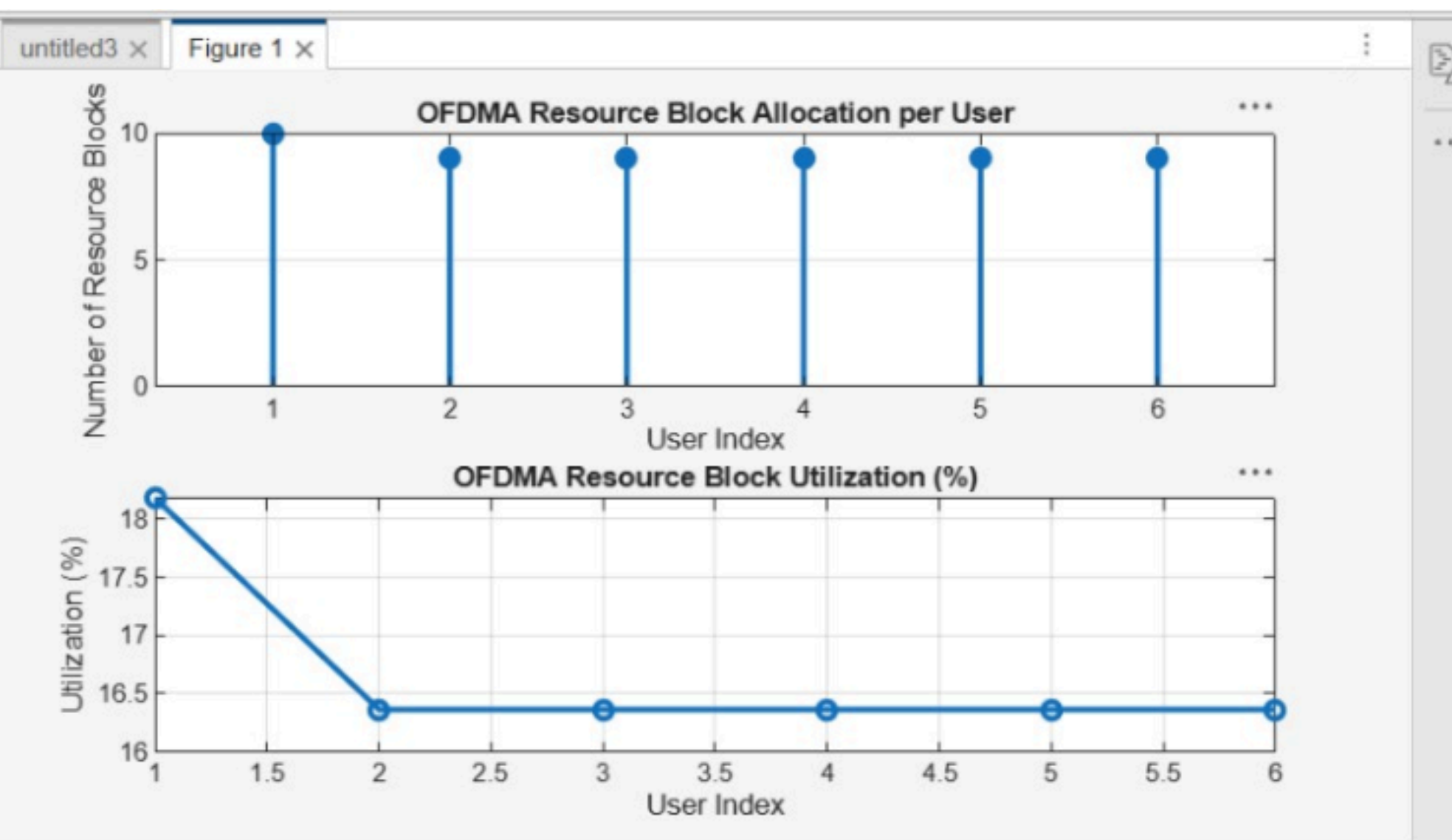
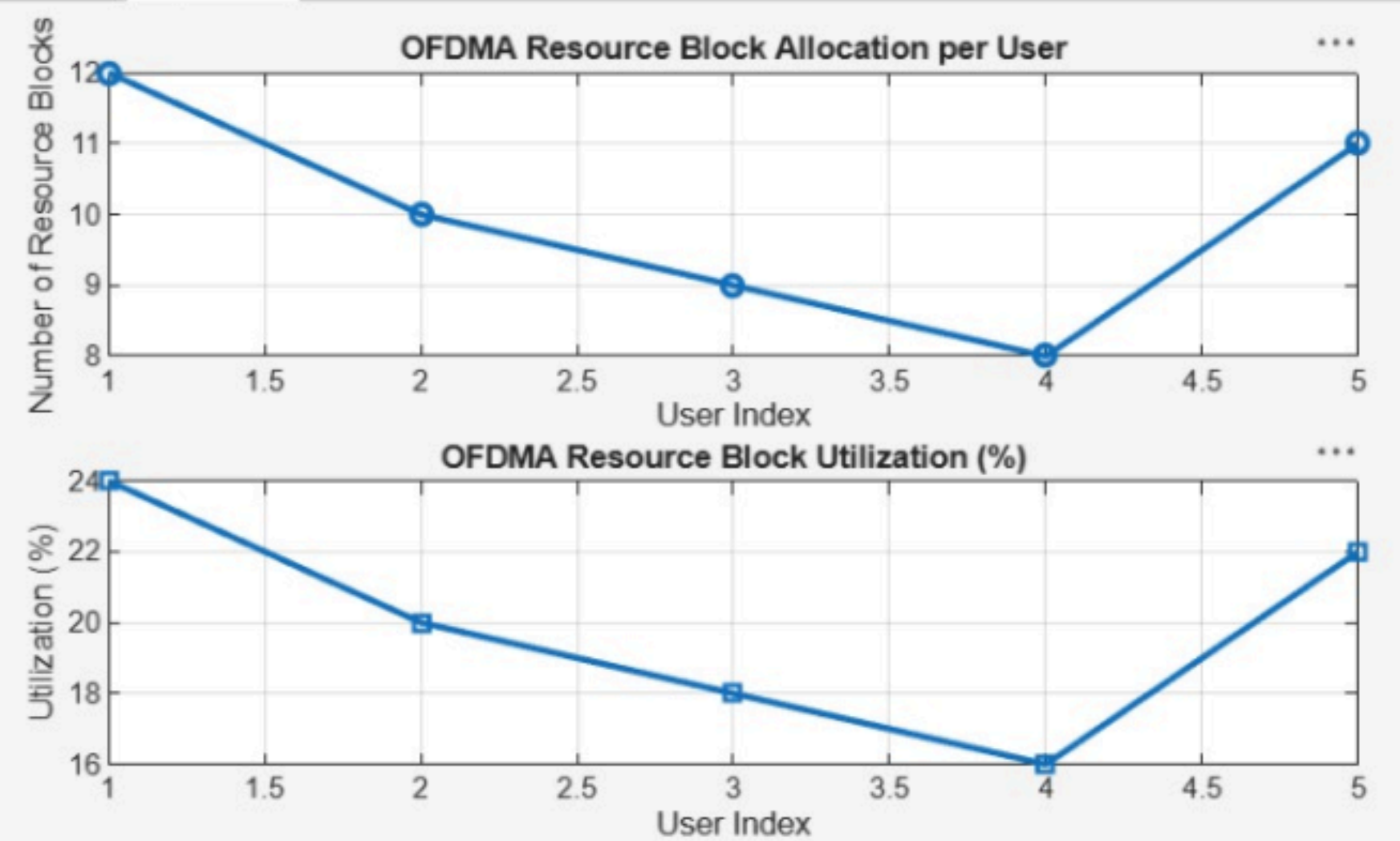


```
1 clc; clear; close all;
2 % System parameters
3 bandwidth = 10e6; % 10 MHz
4 rb_size = 180e3; % 180 kHz per RB
5 % Total Resource Blocks
6 total_RB = floor(bandwidth / rb_size);
7 % Users
8 users = 1:6;
9 % RB allocation (simple equal + remainder distribution)
10 base = floor(total_RB / length(users));
11 alloc = base * ones(1,length(users));
12 alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users)) + 1);
13 utilization = (alloc / total_RB) * 100;
14 disp('Total Resource Blocks:')
15 disp(total_RB)
16 disp('RB Allocation per User:')
17 disp(alloc)
18 figure
19 % ----- Graph 1: RB Allocation per User -----
```



```
untitled4 × untitled31.m × untitled32.m × +
TLAB Drive/untitled32.m
clc; clear; close all;
% Total system resource blocks
total_RB = 50;
% Number of active users
users = 1:5;
% RB allocation to users (example scenario)
alloc = [12 10 9 8 11];
% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;
disp('Resource Block Utilization (%) per User:')
disp(util)
figure
% ----- Graph 1: RB Allocation -----
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% ----- Graph 2: Utilization (%) -----
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```



Command Window

```

clc; clear; close all;
% System parameters
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput)
figure
% ----- Graph 1: Throughput per User -----
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```

User Throughput (Mbps):

4.3200
7.2000
9.7200
5.7600
3.9600

